

Clinical Decision Support for Precision Dosing: Opportunities for Enhanced Equity and Inclusion in Health Care

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Precision dosing aims to tailor doses to individual patients with the goal of improving treatment efficacy and avoiding toxicity. **Clinical decision support software** (CDSS) plays a crucial role in mediating this process, translating knowledge derived from clinical trials and **real-world data** (RWD) into actionable insights for clinicians to use at the point of care. However, **not all patient populations** are **proportionally represented** in clinical trials and other data sources that inform **CDSS tools**, limiting the applicability of these tools for **underrepresented populations**. Here, we review some of the limitations of **existing CDSS tools** and discuss methods for overcoming these gaps. We discuss considerations for **study design and modeling** to create more inclusive CDSS, particularly with an eye toward better incorporation of **biological indicators** in place of race, ethnicity, or sex. We also review **inclusive** practices for collection of these demographic data, during both **study design** and in software user interface design. Because of the role CDSS plays in both recording routine clinical care data and disseminating knowledge derived from data, CDSS presents a promising opportunity to **continuously improve precision dosing algorithms** using RWD to better reflect the diversity of patient populations.

The paradigm of precision dosing arose in recognition that patients vary in how they respond to therapeutics. Moving away from “one-size-fits-all” approaches to strategies that better recognize the variability in patients has improved pharmacokinetic target attainment and efficacy or reduced toxicity in therapeutic areas ranging from **oncology** to **bone marrow conditioning** to **infectious diseases**.^{1–5} Advances in precision dosing have been spurred by the increasing availability and affordability of genetic and other bioassays, advances in statistical modeling and computational tools, and increasingly large data sets made possible by electronic health records (EHRs).⁶ However, the ability to predict how a patient will respond to a particular medication necessitates data and an adequately descriptive model describing how similar patients have responded to that medication.

The precision medicine revolution has not advanced our understanding of medicine in all populations equally. Genomic and pharmacogenomic studies have **overwhelmingly** been conducted in patients of European ancestry.^{7,8} Some patient populations, such as pregnant or paraplegic patients, are regularly excluded from clinical trials and postmarket studies due to difficulties they present in data collection or modeling.^{9–12} Furthermore, models and clinical decision support software (CDSS) often rely on **discretized** characteristics of patients—binarizing sex or gender, or reducing patient ancestry to one of a handful of categories. These simplifications complicate clinical decision making for patients that do not neatly fit within these categories—such as transgender, intersex, or biracial patients.^{13,14} Ironically, given the aims of precision medicine, these simplifications also belie the continuity of physiological

characteristics related to these discrete categories and the high interindividual variability between patients within these categories.

Clinical decision support software occupies a crucial role in bringing precision medicine to the patient bedside (**Figure 1**). It is through these tools that the wealth of knowledge accumulated during drug development and postmarket studies is transformed into actionable insights a clinician can use to find the best course of treatment for their patient. Here, we review the **state of the art in CDSS** with regard to supporting precision dosing in diverse patient populations, and identify solutions for addressing existing gaps. We discuss the challenges in serving historically underrepresented populations throughout the precision medicine data pipeline, from data collection and trial design, to development and interpretation of predictive models, to delivering model predictions to the clinician at the patient bedside.

OPPORTUNITY TO IMPROVE CLINICAL DECISION SUPPORT SYSTEMS TO ADDRESS PATIENT DIVERSITY

Clinical decision support software encompasses a variety of software tools that supply clinicians with information for consideration during their treatment decisions. In the precision dosing field, this information may include drug–drug interaction or pharmacogenomic (PGx) alerts, or predicted drug exposures or the likelihood of treatment efficacy or drug toxicity associated with a prescribed regimen (**Figure 1**). These algorithms and models, which may be based on statistical analysis of data, machine learning or artificial intelligence, are typically individualized to the patient based on their medical history, their laboratory or

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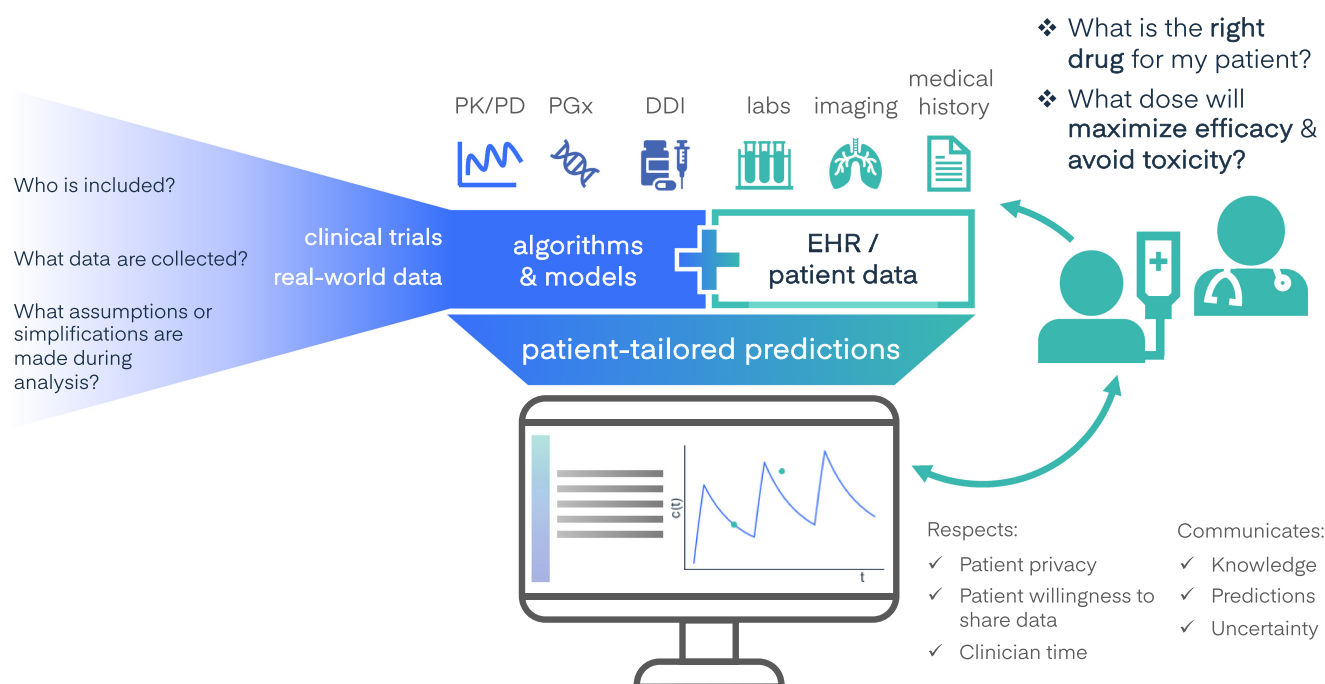


Figure 1 CDSS (Clinical decision support software) occupies a crucial position in precision medicine, translating the data collected from clinical trials and real-world data, such as those describing PK/PD (pharmacokinetics/pharmacodynamics), PGx (pharmacogenomics), and DDIs (drug–drug interactions), into actionable insights at the point of care. CDSS combines these models and algorithms with data about a specific individual, tailoring predictions to the patient.

imaging results, and/or their genomic data. While developers of such algorithms may have the best intentions in avoiding biasing the algorithm by race, gender, or other classifications, there is still considerable risk involved in applying algorithms at the point of care.

Case study: Race, kidney function, and clinical decision making

In clinical pharmacology, renal function is typically assessed using estimated glomerular filtration rate (eGFR). While some early equations like the Cockcroft-Gault equation do not include race, the newer and more accurate 2009 Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) and the 1999 Modification of Diet in Renal Disease Study (MDRD) equations do include race as a factor. The race factor in these equations is intended to correct for the relatively higher serum creatinine levels measured in Black vs. White individuals with otherwise equal physiology, which was statistically identified in the data sets that formed the basis for those equations. However, close scrutiny of the underlying data set shows that the data sets were unbalanced in various ways between race groups (e.g., kidney disease type, socio-economic status).¹⁴ Additionally, race cannot be considered an accurate classifier for the individual, as racial groups are **genetically and socially highly heterogeneous**. The resulting problem with the inclusion of race in estimators of renal function is that it falsely classifies Black individuals to a lower chronic kidney disease risk score and reduces their access to certain drug treatment or other specialty health care, thereby exacerbating existing racial

disparities.¹⁵ A positive development is that a new GFR equation has recently been presented by the CKD-EPI consortium that **does not include race** as a factor.¹⁶

Unfortunately, however, the solution is not always as simple as just removing or ignoring race or gender as a factor in the algorithm. For example, a retrospective analysis of National Cancer Institute data found that removing race from the CKD-EPI GFR equation would have excluded 67–163% more Black patients from receiving anticancer therapy due to lower estimated renal clearance compared with the standard CKD-EPI equation.¹⁷ In another example, it was demonstrated that an algorithmic risk score predicting general healthcare risks exacerbated racial disparities in access to health care, even though it did not include race as a factor itself. The reason was that the risk model was trained on healthcare cost data, which acted as a proxy for race in the model and steered Black patients away from care they would have received if White.¹⁸

Addressing limitations of existing clinical decision support tools

In 2020, a request for information addressed to professional medical societies and other stakeholders was sent out by the US government to survey perspectives on the race-related health disparities in relation to CDSS, signaling that these issues are recognized at the highest levels.¹⁹ Recommendations from parsing the feedback on the request for information focused on acknowledgement and awareness of the issue, appropriate communication, improved clinician training, and re-evaluation of the use of race in clinical algorithms.

Vyas and colleagues, reviewing clinical algorithms that historically have led to bias in clinical care, recommended three questions that clinicians should ask when developing or implementing race as a predictor:²⁰

1. Is the need for race correction based on robust evidence and statistical analyses?
2. Is there a plausible causal mechanism for the racial difference that justifies the race corrections?
3. Would implementing this race correction relieve or exacerbate health inequities?

We feel these questions similarly apply to questions of gender, or other subgroups for which disparities in health or access to health care exist.

Arguably the main purpose of CDSS is to reduce uncertainty that a clinician has about the diagnosis or treatment for a specific patient. However, not every CDSS application provides a clear estimate of the remaining uncertainties in its predictions or advice, and in what ways the predictors contribute to predictions and reduce uncertainty. Additionally, much simpler techniques that do not use any algorithm should be considered, such as nudges that can be built into a CDSS tool to make users aware of existing or historical biases. An example of this was the implementation of a race-dependent warning message into an EHR at two large academic centers to highlight previous disparities in the admission process to specialty cardiology care.²¹

Algorithms in CDSS are often proprietary and not available for close scrutiny, and clinical decision support vendors may not always seek publication in peer reviewed journal articles. “Black box” models, where the rationale for a particular prediction cannot be audited, are particularly susceptible to building in existing social biases. Because pharmacokinetic/pharmacodynamic (PK/PD) models clearly describe the mathematical impact of each predictor on a prediction, this form of model is closer to a “glass box” model; however there is a growing toolkit for bringing explainable deep learning models to healthcare applications.²²

It is interesting to note that the European Union’s General Data Protection Regulation provides several protections in this context: firstly it allows a citizen, if affected by a decision produced by machine learning/artificial intelligence (ML/AI), to know the reasons behind that decision. Additionally, Article 22 of the General Data Protection Regulation stipulates that individuals can choose not to be the subject of such a decision, as they have the right of “meaningful human review.”^{23,24} However, it is still unclear how applicable, enforceable, and beneficial these articles are for CDSS, since CDSS tools often only *support* a decision that is ultimately taken by a clinician and are not truly autonomous.²³ While such regulations may offer some protection for European Union patients, lack of clarity regarding the regulation and added implementation burdens may impede the availability of ML/AI algorithms in the European Union and deprive its citizens of state-of-art diagnosis and treatment. Federated learning approaches have been proposed as one way to train models with remotely hosted data sets and without the need to accumulate data centrally.^{25,26}

COLLECTION AND ANALYSIS OF REAL-WORLD CLINICAL DATA TO SUPPORT EQUITY IN OUTCOMES

Although there is a growing recognition for the need for diverse clinical trial populations,²⁷ drug development trials typically do not reflect the diversity of the broader patient population.²⁸ CDSS tools for precision dosing relying only on data collected through these avenues are therefore inherently limited. Given that clinical trials may represent < 50% of the patient diversity present in the real-world market population,²⁹ availability of real-world clinical data (RWD) provides an opportunity to understand drug benefit and risk beyond what is possible using data from registrational trials (Figure 2). The collection of RWD, including administrative/insurance claims data, patients’ EHRs, disease/product registries, and genomic data, will help improve our understanding of drug response in underrepresented patient populations and enable clinicians to select the right medication at the right quantity and frequency for their patient.

There are a growing number of examples of RWD being leveraged to continuously improve drug dosing recommendations after the treatment has been approved.^{30–34} In one study, differences in enoxaparin dosing and disposition were evaluated in children with obesity using the Pediatric Trials Network data repository. The study found that obesity was a significant predictor of anti-Xa concentration in children and recommended using fat-free mass (FFM) instead of total body weight for weight-based dosing.³⁵

Real-world data can be collected from postmarketing surveillance studies conducted across multiple centers or directly from the EHR systems. CDSS can also be leveraged to collect

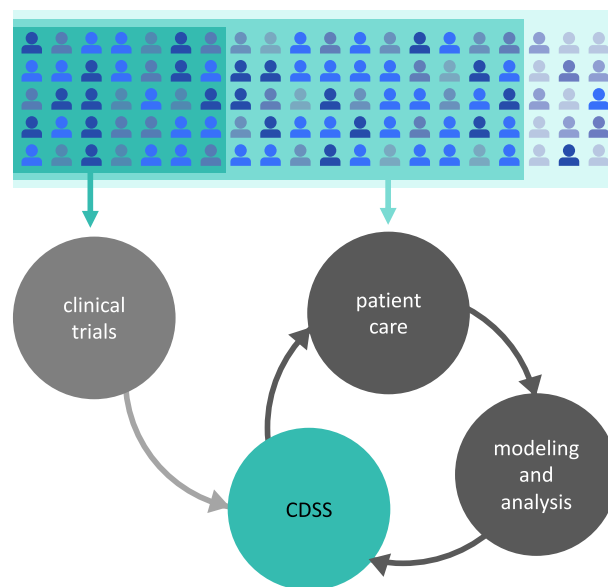


Figure 2 CDSS (clinical decision support software) translates findings from clinical trials to inform patient care. Since clinical trials may reflect less than half of the diversity of the broader population, real-world data present an opportunity to continuously improve CDSS over time to better reflect the variability between patients. However, some individuals may not have full access to medical care, and could still be underrepresented in CDSS.

RWD from EHRs and thereby implement a continually learning system.^{25,36,37} For example, dosing decision support tools integrated within EHRs can collect additional clinical and demographic data that ultimately inform dosing strategies in underrepresented subgroups. Current regulatory frameworks in the United States and Europe are not well prepared to accommodate such self-learning systems,³⁸ although the US Food and Drug Administration (FDA) has recently released a comprehensive overview of draft approval strategies for developing ML/AI systems.³⁹ A shift to a more holistic regulatory view of how AI/ML systems are implemented and used in the clinic will likely be required to unlock the full potential of such systems while ensuring their safety and efficacy.⁴⁰

Historically, individual patient data on gender has not been collected in the general patient population. One reason for this is that, even if an individual is willing to reveal their gender to a provider, EHRs in many health organizations have been unable to store and share this information appropriately.⁴¹ Additionally, the support for storage of nonbinary genders has varied considerably from system to system, rendering gender data invisible in aggregate health data sets and leading to widespread inaccuracies in their use in EHRs.⁴² For race and ethnicity, similar data storage and interoperability problems exist. Electronic systems or paper forms that record race typically use the categories recommended by the Office of Management and Budget (OMB), which consists of just five racial classes and two ethnicities. These categorizations are based on historic and outdated understandings of racial differences and were not intended for describing underlying biological differences.^{43,44}

Despite the promise of RWD to improve drug dosing in underrepresented patient subgroups, there are still several hurdles standing in the way of collecting RWD to update drug labels and dosing guidelines. Interoperability of EHRs requires improvements to allow for a full picture of a patient's clinical profile and treatment trajectory. Recent regulatory policies such as the Prescription Drug User Fee Act VI and the 21st Century Cures Act have begun to close gaps in data interoperability, quality, and accessibility,^{45,46} and several private and public organizations, such as Flatiron and Komodo Health, have formed to facilitate RWD aggregation.⁴⁷ Another challenge is that some patients, typically those most historically underrepresented in medical research, are reluctant to share information, potentially due to past experiences of discrimination or the poor historical track record regarding equity in clinical care and research.⁴⁸ The lack of information sharing can negatively impact the quality of data in a patient's medical record, which has consequences for future medical decision making.⁴⁹ Finally, the level of evidence required to leverage RWD to inform drug development and update drug labels is unclear. The FDA recently created a framework for evaluating the potential use of real-world evidence to help support the approval of a new indication for a drug or to help support or satisfy drug postapproval study requirements.⁵⁰

Pharmaceutical companies, healthcare providers, and regulatory agencies together play an important role in enabling the collection of RWD to improve dose recommendations in underrepresented patient populations. Pharmaceutical companies will need

to expand postmarketing surveillance efforts with phase IV studies that include patient data in underrepresented patient groups. Provider organizations should continue to work with EHR vendors to improve the diversity, accuracy, and reliability of patient data. Additionally, the FDA must continue to evolve their best practice guidelines on leveraging RWD to update drug labels and dosing guidelines. Finally, modelers and statisticians must analyze RWD and incorporate it into predictive models with a thorough understanding of the biases and limitations of their data.

MODEL DEVELOPMENT TO SUPPORT CLINICAL DECISION MAKING IN DIVERSE POPULATIONS

Clinical trial data and RWD are typically transformed for use in precision dosing CDSS in two formats: in PGx platforms, which summarize possible differences in medication metabolism, transport, and drug effect based on a patient's genetic profile, or via predictive models, which forecast how a patient will respond to a particular treatment regimen given patient descriptors, biomarkers and/or genetic information. Design considerations for delivery of PGx information have been recently reviewed.⁵¹ Here, we focus on the development of predictive models for use in CDSS.

Models, whether based on PK/PD principles or developed using machine learning approaches, often incorporate patient sex, race, or ethnicity as predictive features. These demographic data act as proxies for real underlying differences in physiology or pharmacology observed between patients, such as genetically inherited differences in metabolic enzymes or sex-related differences in drug bioavailability.^{13,52,53} Such predictors can be implemented directly, as covariates acting on a model parameter, such as a difference in clearance between male and female patients. These predictors can also be implemented indirectly through an intermediate covariate that relies on these data as input variables, such as eGFR or estimated FFM. The ubiquity and ease of collection of demographic data make them convenient for inclusion as model predictors, thereby improving the model's predictive performance and increasing the likelihood of a positive treatment outcome for the patient. On the other hand, the specific use of sex, race, and ethnicity in PK/PD modeling and precision dosing also risks exacerbating existing health disparities.²⁰

An additional issue is that treating variables like sex and race as **discrete categories** masks the high physiological variability between individuals *within* these categories and complicates clinical decision making for patients that are intersex, transgender, biracial, or do not provide information on race or ethnicity.^{13,14} As one example of overgeneralization of the impact of race, the drug label for carbamazepine indicated genetic screening for patients of Asian ancestry due to adverse drug events related to the human leukocyte antigen (HLA) B*1502 allele. However, these findings were based on data collected from just **two Asian countries** and a large study of nearly 29,000 patients suggested that screening based on ethnicity assumptions may **miss a high number of at-risk individuals**.^{44,54}

In contrast to model development during drug development, model development for precision dosing should optimize for forecasting accuracy, rather than descriptive ability, and should consider the availability and suitability of covariates for clinical implementation. Here, we discuss recommendations for translation

Table 1 Summary of recommendations for developing inclusive models for precision dosing

Model development stage	Recommendation
Study design	
Sample size, inclusion/exclusion criteria	• if pharmacogenetic effects are suspected, ensure sample size adequately allows covariate estimation based on expected population frequencies • include patients with a wide range in body size, disease severity, and comorbidities • globally recruit patients
Data collection	• collect PGx data, where feasible • allow patients to decline to respond to questions about sex/gender identity and race/ethnicity
Model development	
Covariate model selection	• Consider physiological mechanism, e.g.: - a include gender or hormone status as covariates for absorption rate - b include fat-free mass instead of total body weight and sex as covariates on volume of distribution - c Prioritize PGx data (where available) over race/ethnicity • Consider clinical relevance of effect sizes • Where race/ethnicity varies from site-to-site in a multi-institution study, assess whether identified race effects could be batch/site-specific effects
Model reporting	• in addition to the final model, report parameter estimates for the model without sex/race covariates
Model evaluation	
Subpopulation assessment	• Evaluate model predictive performance iteratively, using tools like PsN proseval • Evaluate models on a holdout patient population, where feasible • Evaluate bias and error across different races and sex/genders

PGx, pharmacogenomic. PsN, Perl-speaks-NONMEM.

of models to the clinic with an eye toward description of race/ethnicity-related covariates and sex/gender-related covariates, as summarized in [Table 1](#).

Data collection and study design

Models are only as good as the **data used to develop them**, and so the design of studies with an aim of building predictive models should consider covariate specification and patient diversity during data collection. The model development population should align with the **intended precision dosing population**.³⁶ Clinical trials are overwhelmingly conducted in the United States and other highly economically developed countries,⁵⁵ and global recruitment could serve to address this **inequity** and improve our understanding of interpatient variability in populations not well represented in these countries. Global recruitment of patients may also allow for more rapid and potentially more cost-effective means of obtaining diverse patient populations.⁵⁵ However, global conduct of clinical research studies can also worsen global inequality.⁴⁸ For example, one study found that 40% and 60% of clinical trials conducted in India and South Africa, respectively, led to market authorization in the European Union or the United States without a new drug application approval in India or South Africa.⁵⁶ This gap exists despite the World Medical Association Declaration of Helsinki emphasizing that research conducted in a vulnerable patient population, which undoubtedly includes patients without access to health care outside a clinical trial,⁵⁷ is ethical only if the population in which the research is conducted **stands to benefit from the research**.⁵⁸ With the rise in studies outside of the European Union and the United States, genomic data catalogs show increasing ancestral diversity. However, these gains are principally caused

by intensifying research efforts in countries in East Asia, and so patients of African, South Asian, and/or Hispanic/Latino ancestry continue to be **underrepresented**.^{8,59}

As race and ethnicity serve as proxies for underlying pharmacogenetic differences, for drugs in which a pharmacogenetic component is suspected, the gold standard should be to collect pharmacogenetic information. Where this is not feasible, race/ethnicity data should be patient-reported,^{60,61} and allow for patients to select multiple options or to refuse to respond. Collecting more detailed categories than those defined by OMB, such as different Asian ancestries, may better approximate pharmacogenetic differences.^{43,62} However, lack of standardization of these categories can complicate comparison across studies or implementation into clinical practice.⁴³ Consistent forms should be used throughout the study period, since questionnaire differences can impact responses.⁶³ Diverse patients should be recruited from each participating location, allowing differentiation between study-specific or site-specific effects and race/ethnicity effects.

Patient sex and gender should be collected separately, with options to decline to answer.⁶⁴ Since some PK variability may be explained by hormones rather than chromosomal sex, hormone replacement therapy comedication information, such as duration on therapy, should be collected.^{13,65,66}

Sample sizes should be calculated to allow for adequately powered estimation of effect size.⁶⁷ Underrepresentation of groups within a model-building cohort can lead to models that perform poorly in these subpopulations.⁶⁸

Many vulnerable patient subpopulations, such as pregnant patients, are routinely excluded from study design out of ethical concerns.¹¹ Other populations, such as patients with amputated limbs,

paraplegic patients or patients on renal replacement therapy, may be excluded from pharmacokinetic studies due to the challenges they present for pharmacokinetic modeling. However, drugs benefitting from precision dosing are regularly prescribed to these patient populations and are therefore prevalent in RWD, and there is therefore a need to better study these subpopulations (Figure 2).

Model development

During model development, only physiologically plausible relationships between covariates and PK parameters should be considered, and covariate models with the most explanatory power should be selected rather than blindly pursuing statistical significance. Some journals, such as the *American Family Physician*, do not accept manuscripts whose findings indicate biological differences by race without further justification.¹⁹ While this decision is praiseworthy and should be adopted by other journals, it is only a partial fix. As we described above, even if race or gender is not explicitly included as a factor, other factors may act as a proxy.¹⁸ Covariate interaction should also be considered. For example, in one pharmacogenomic study of efavirenz, Black race remained a statistically significant covariate in addition to genotype until gene–gene interactions were incorporated.⁶⁹ Close collaboration between clinicians and statisticians is vital for selecting appropriate covariate models. Model developers should consider not only statistical significance but also clinical significance when choosing to report a drug's PK/PD as impacted by race or sex; an effect size of 5% is unlikely to meaningfully impact decision making but could reify perceptions of the differences between individuals of these categories.

One difficulty, both in building predictive models and in applying predictive models, is how to handle missing or uninformative data.⁷⁰ Sometimes, patients for whom reported race or ethnicity is unavailable or uninformative are categorized as part of the majority ethnicity or grouped together with other ethnicities to ameliorate low sample frequencies.⁷¹ This practice should be avoided; it is not grounded in physiologically plausible relationships, it could attenuate effects sizes, and it could reinforce systemic biases in health care in which some races or ethnicities are considered “default.”

We recommend instead that authors report both a model without genotype, race/ethnicity or gender/sex, as well as the final version of the model with all covariates. This approach provides several advantages. First, patients for whom reported race/ethnicity and/or sex/gender is missing or uninformative can inform the covariate-free version of the model and be removed from the final version of the model, avoiding the previously mentioned data missingness issues. Second, the covariate-free model can be applied to patients for whom this demographic information is unknown—whether because the patient declined to provide it, or because the patient presents to the clinic as a John/Jane Doe and this information is not yet available—without assuming values for these covariates. Covariate-free models are typically trained as part of a standard population PK model development workflow, so this is a negligible burden to model developers. Another option could be to incorporate these covariates using a mixture model, which allows data from patients for whom data are missing or uninformative to be used; however, this approach is more computationally complex

and may be less suited for situations where data are missing not at random.⁷⁰

Case study: Covariate model informed by knowledge of physiology

As a narrative example, a statistician notices that model diagnostics suggest patient sex improves model fit for an oral medication. Consulting with a clinical pharmacologist, they create a set of mechanistic covariate models to test. Bioavailability and absorption are modeled as depending on patient gender and time on hormone replacement therapy, since estrogen impacts gut transit time.⁶⁶ Because the drug is hydrophilic, distribution volume is modeled as dependent on FFM or lean body mass instead of as an interaction between total body weight and sex.¹³ For a lipophilic drug, fat mass could have improved the fit instead. As the drug is known to be metabolized by cytochrome P450 2C19 (CYP2C19), a cytochrome influenced by estrogen therapy,¹³ clearance is modeled as dependent on gender and hormone comedication. The resulting model uses predictors like gender and hormone comedication, and therefore a clinician can apply the model at the point of care to treat their transgender patient with less ambiguity.

Model evaluation

Models intended for application to clinical practice should be evaluated for their predictive performance. Iterative prediction of subsequent drug or biomarker concentration, such as through Perl-speaks-NONMEM proseval,⁷² more closely mimics the performance of models in the clinic³⁷ compared with approaches that use all available data for a patient to predict previous concentrations. Even if demographic covariates are not found to be informative of PK parameter estimates, error or bias in model predictive performance should be evaluated across these data categories to ensure the model does not worsen existing health disparities.

USER INTERFACE DESIGN TO SUPPORT CLINICAL DECISION MAKING IN DIVERSE POPULATIONS

All of the above considerations have implications for the architecture and design of CDSS for precision dosing. Developers of these systems must create interfaces that reflect patient diversity and gracefully handle missing data. To be adopted by clinicians and improve clinical practice, they must also be fast, be integrated into clinicians' workflows, and provide clear recommendations and guidance.^{73–75}

There are numerous challenges for building a system that meets these criteria. Clinicians are already overburdened by time-consuming clerical work, so developers must design usable systems that do not add unnecessarily to that burden.^{76,77} Many providers lack specific training in healthcare issues for vulnerable populations, leading to disparities in health outcomes.^{78–80} It is therefore important that the CDSS prompt for the appropriate data while minimizing redundancy and maintaining a streamlined user interface. Poor user experience and interface design can impair clinicians' cognitive performance and lead to catastrophic consequences for patient care.^{81–83} We propose several guidelines

for designing user interfaces to reflect patient heterogeneity and underrepresented groups in CDSS for precision dosing.

These tools should support flexible options for inputting demographic data such as race, ethnicity, sex, and gender. A lack of flexibility in medical information systems contributes to patient erasure and can prevent patients from receiving appropriate care.^{84,85} Systems should support multiple selections and write-in options for these demographic fields. Scheuerman and colleagues presented several options for gender input fields in a survey, and found that most participants considered a free-text field to be the most inclusive option in healthcare settings.⁸⁶ Though it was not an option within the survey, many participants preferred medical forms that asked for pronouns and anatomical details, rather than asking for the patient's gender identity. This latter option could provide information that is both more structured and useful for modeling as well as more informative than free-text gender or sex fields.

Flexibility also includes allowing missing data in any or all of these fields, as patients from marginalized groups may feel unwilling to disclose personal information to providers.⁴⁹ However, if these data are required for model predictions, either the patient would be excluded from precision dosing, or the software would need to use a default value for race, gender, or other demographic variables, which in its own way constitutes a normative judgment. This stresses again why it is important for models to be reported both with and without demographic covariates: this way, CDSS can avoid assigning a default race or sex to patients when their demographic information is unknown.

As described above, demographic information is often used as a proxy for physiological or anatomical data, such as hormone levels or FFM. We recommend supporting these physiological variables instead of or in addition to demographic ones. Rich, clinically relevant patient information provides a clearer overall picture of the patient's health and needs.^{74,87} If the system prompts for physiological data, it should provide clear and succinct guidance to the user on what to input, with links to relevant evidence.⁷⁴

Lastly, we recommend that developers of CDSS for dosing take an iterative and participatory approach to designing systems. By involving stakeholders in the design process, they will develop solutions that best suit users' needs and are more likely to be adopted and used correctly. This participatory approach should involve interviews and usability testing with clinicians. Formal usability testing and analysis with real end users can improve the quality and usefulness of a system.⁸⁸ Though patients themselves are not typically the end users of CDSS, their expertise (especially that of marginalized groups and organized patient advocacy organizations) can also inform CDSS developers. As an example, transgender people have compiled their own resources to understanding dosing regimens in the absence of clear clinical guidelines.⁸⁹ Developers of CDSS could provide a great deal of value to these patient populations, and improve their software's utility, by working with them to understand their needs from a dosing tool.

CONCLUSIONS

Clinical decision support software forms the interface between medical professionals and the accumulated knowledge of how

patients respond to drugs. This accumulated knowledge does not yet represent the full diversity of patients who seek medical treatment, limiting the ability of precision dosing to tailor regimens to all patients. RWD should be leveraged to improve drug dosing in underrepresented patient populations starting from drug development to post approval. During drug development, RWD can help inform important clinical pharmacology questions such as dose escalation, covariate analysis, exposure–response, and dose optimization. Following drug approval, RWD should continue to address gaps and questions related to safety, efficacy, and exposure–response of a treatment in the postmarketing phase. Finally, for a subset of drugs where predictive models are routinely used to optimize the dose for patients within a hospital setting, CDSS could serve as an effective approach to collect diverse patient data types directly from the EHR. Data collection through CDSS and communication of these insights in a way that both accurately captures patient diversity but is conducive to data analysis and modeling is therefore of the utmost importance.

In this review, we have focused on how collection of clinical trial data and RWD, software design, and modeling can be approached to enhance equity in health care. However, ensuring the benefits of precision dosing accrue to all individuals (Figure 2) will also require coordination between public health institutions, healthcare institutions and professional societies, and payers.

Neither precision dosing nor equitable health care allow for “set it and forget it” implementation. Precision dosing models and PGx databases should be continuously updated as additional data become available.^{40,50} Models and predictions should similarly be continuously evaluated, particularly to ensure they help to reduce healthcare disparities and not exacerbate them. This feedback loop, of using CDSS at the point of care to inform patient care, and then using patient data collected through CDSS to update and inform CDSS, holds particular promise for patients historically underrepresented in clinical trials (Figure 2). Equitable implementation of this continuous learning process requires careful attention throughout the study design, data collection, and modeling process.

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